

Mineral Chemistry, Fluid Inclusion Studies and P-T Path of Gneiss-Granulite Rocks of Schirmacher Region, East Antarctica

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Abstract

In the Proterozoic complex of Schirmacher region of East Antarctica a retrograde P-T history has been inferred through quantitative geothermobarometry and fluid inclusion studies of the felsic gneisses and mafic granulites.

The temperature and pressure of metamorphism has been estimated through different models of geothermometers and geobarometers. The P-T results suggest two major episodes of metamorphism, one recording earlier high P-T granulite facies metamorphism and the other latter retrograde upper amphibolite facies metamorphism. The gneisses of the region are formed during the amphibolite facies metamorphism.

The fluid inclusion studies of felsic rocks showed mainly three stages of fluid entrapment, the early stage monophasic CO₂ inclusions, followed by monophasic carbonic inclusions and the early carbonic aqueous inclusions in trails, and lastly by late carbonic aqueous and biphasic aqueous inclusions in trails. The mafic granulites also record three phases of fluid entrapment in quartz and garnet minerals, viz. earliest pure CO₂ inclusions, CO₂-H₂O inclusions and aqueous inclusions.

A good correlation between the fluid and mineral data is observed. The isochores typical for high density CO₂ fluids fall well within the P-T box estimated by mineral thermobarometry. On the basis of fluid inclusions, a pervasive influx of deep-seated CO₂ rich fluid from the mantle is envisaged in the formation of the granulites. The subsequent stages showed decrease in CO₂ density and progressive influx of hydrous fluids implying a decompressional P-T-t path for the generation of retrograde amphibolite facies metamorphism in the area.

Introduction

The challenging problems in modern geology is to understand the nature and cause of deep crustal metamorphism, particularly granulite facies metamorphism, and the processes that have operated thereon. Mineral P-T estimates and CO₂-rich fluid inclusions help us to know the role of physical conditions

of metamorphism and how the granulite facies assemblage developed at deeper crustal levels. Presence of relict minerals provide information on changes in P-T conditions during the metamorphic history of the rocks. The mineral zoning studies have yielded different P-T paths in amphibolite-granulite terrains. The fluid inclusions and P-T evolution of the felsic gneisses and mafic granulites in Schirmacher region, East Antarctica (between latitudes $70^{\circ}44'30''\text{S}$ to $70^{\circ}46'30''\text{S}$ and longitudes $11^{\circ}22'40''\text{E}$ to $11^{\circ}54'00''\text{E}$) are considered here in order to understand their evolution.

Field Geology

The field geology, the structural details along with the description of the various rock types and geological map of the Schirmacher region, East Antarctica are given by Sengupta (1986, 1988, 1991, 1993) and Stackebrandt, *et al.* (1988). A brief description is, however, given here.

The gneisses constitute more than 85% of the exposed bed rocks in the region. The other important lithological units are mafic granulites, charnockites, metapelites, calc-silicate rocks and amphibolites. Besides these, there occurs dyke rocks such as dolerites, lamprophyres, microgabbros, pegmatites and aplites as well as tectonites such as blastomylonites. Agmatic structures engulfing basic relics are also seen in the gneisses.

On the basis of nature and composition, the gneisses are being classified into biotite-gneisses, garnet-biotite gneisses, hornblende gneisses, augen gneisses, and leuco gneisses. These felsic units occur as stringers, porphyroblastic stringers of augen gneisses, bands and as lenses of amphibolite facies, within the granulite facies rocks, the latter showing retrogression to the amphibolite facies. In general, the gneisses show a northwesterly trend with minor variation in strike from place to place with foliation planes gently dipping towards SW. Veins of pegmatite with variable thickness ranging from few inches to few feet are injected into the gneisses. The pegmatitic veins show both conformable and cross-cutting relationships with the foliation of the gneissic rocks.

The mafic granulites are characterized by black to greyish black colour, the colouring is due to pyroxenes and feldspars. The rock is generally medium grained and are intercalated with pyroxene gneisses and felsic gneisses showing sharp to gradational contacts. They occur as sills parallel to the structure of the gneisses or as lenses or as boudinage structures. Foliation is usually not so distinct. Compositely layered mafic-felsic gneisses traversed by pegmatites are also observed. Rarely these rocks are found as irregular shaped bodies within felsic gneisses. In central Schirmacher, one can also observe distinct bands of

mafic granulites being folded along with metapelites and calc-silicate rocks. Most of the amphibolites of the region are developed by retrogression of mafic granulites, invariably the transformation is associated with injection of the felsic or pegmatic material.

The mineral assemblage of the rocks in the Schirmacher region are generally those typical of medium P-granulites, *i.e.*, orthopyroxene-clinopyroxene-plagioclase-garnet-hornblende in mafic rocks, orthopyroxene-garnet-K-feldspar-plagioclase-quartz in orthopyroxene gneisses and K-feldspar-plagioclase-quartz-biotite $\pm$ garnet $\pm$ hornblende in felsic and intermediate gneisses, while pelitic rock are consistent with the stability of biotite-sillimanite-garnet $\pm$ graphite assemblage.

The early granulite facies mineral assemblages observed in Schirmacher region, are characterized by the pyroxene granulite facies rocks. The subsequent overprinting amphibolite facies assemblages are characterized by hornblende and biotite and their formation is attributed to hydration of mafic minerals and recrystallization. Relicts of early granulitic assemblage are frequently preserved in mafic granulites as evident from the instances of small relict grains of orthopyroxene and clinopyroxene within amphiboles. Thus, the later amphibolite facies assemblage must have crystallized at temperatures substantially lower than the early granulitic assemblage (Rameshwar Rao and Sharma, 1995).

Petrography

Felsic Rocks of the Region

The rocks show inequigranular, interlocking to porphyroclastic to granular textures. The textural variation is due to the varying intensities of deformation. The major minerals quartz, K-feldspars and plagioclase occur in various proportions, having sutured boundaries and interlocking textures. The other important minerals of these rocks being biotite and $\pm$ garnets. The accessories mainly include titanite, ilmenite, zircon and muscovite. In general, the phases have an exceptionally fresh texture, except for minor secondary alterations of chloritization and sericitization shown by biotite and plagioclase minerals.

Quartz except in high strain zones do not show much flattening. Quartz grains exhibit extensive undulose extinction, development of subgrains and recrystallized grains indicating dynamic recrystallization via subgrain rotation and grain boundary migration. The highly elongated quartz ribbons with large aspect ratio in some quartzo-feldspathic mylonite indicate, metadynamic recrystallization at high rates of hot deformation (Sengupta, 1988).

K-feldspars are mainly microcline and microcline-microperthite as well as early untwinned microperthite. The early K-feldspar are medium grained porphyroclasts contain inclusions of plagioclase and quartz, formed as megacrysts during the crystallization of the granite. The predominant microcline-microperthite and microcline with quartz and fine grained plagioclase must have crystallized during late stages. These K-feldspars replace the early formed K-feldspars and plagioclases. The K-feldspar invariably show uneven undulose extinction and minor development of subgrains and recrystallized grains. The serrated boundaries and recrystallized grains indicate that they were deformed by recrystallization accommodated dislocation creep.

The gneisses, in general, also contain a fair amount of perthites with or without microcline cross-hatched twinning. Perthites are mainly of lamellae with uniform thickness, as well as discontinuous lamellae. Fine and coarse lamellae are present within the same grains. The other variety is of elongate bleb type and some of the blebs show sigmoidal shapes. Apart from these, the vein perthite and patch perthites are also common in these rocks. In case of blebs they show minor alterations indicating calcium rich nature. The perthite development can be related to deformation. Even though these gneisses show strong high temperature ductile deformation, there is no definite microstructural evidence of different perthite types related to deformation. The perthitic K-feldspars of these gneisses are highly modified due to ductile deformation and hence to deduce mechanisms of their development is difficult. Nonetheless, their different shapes and coarseness may be due to high temperature exsolution, as well as due to water activity.

Plagioclase occur in three varieties: (i) as inclusions in early K-feldspar and rarely in garnet, (ii) as medium to coarse grained porphyroclasts with serrated boundaries and (iii) as myrmekitic plagioclase around the K-feldspar margins. In general, the plagioclase show sericitization. Apart from polysynthetic growth twins, number of plagioclase grains show twins which are thin, end abruptly within the grain. They are commonly bent and appear to be mechanically produced. Some of the plagioclase grains are divided into three or more grains with misorientation and deformational twinning while others with strong undulose extinctions and bent twin lamellae show subgrains and recrystallized grains. Even though the feldspars show minor intragranular cracking without displacement, the majority of the cracks are transgranular which could be produced much later.

Garnet is present in some of the quartzo-feldspathic gneisses, mostly associated with biotite. The small biotite flakes surrounding the garnet appear to be replacing the garnet along the margins and along the fractures in garnet. In some gneisses the garnet clots are sporadically distributed and impart

foliation. In some cases the replacement is advanced producing atoll garnets. Some of the small garnets having quartz inclusions at the margins in symplectic intergrowth indicate that, these garnets appear to have developed or grown over the biotite. This texture indicates that garnet was stable or developed during the high temperature ductile deformation.

The garnets having early quartz grains as inclusions and surrounded by well developed recrystallized quartz grains, indicate that these garnets have been developed by nucleation and crystal growth at the curved boundaries of quartz grains which produce hollow monocrystals of quartz. These textures and other textural characters indicate that garnet grew in the granite melt and some of the new garnets were stable during deformation.

Muscovite which is minor mineral in these rocks occurs interlayered with biotite at places, this is used as textural evidence for considering muscovite as primary. At some place muscovite appears to be of secondary nature also. Biotite mainly occurs along with garnet in clusters with different orientations, as well as isolated laths defining foliation. In general, the biotites were stable during deformation. However, minor rare patches of chlorite occur in biotites which is due to autometamorphic alteration by late stage fluids and in this case, the biotite alteration is due to late stage melt crystallizing quartz and K-feldspar.

Mafic Rocks of the Region

The mineral assemblage of the mafic granulite rocks is somewhat variable. They are mainly composed of orthopyroxene, clinopyroxene, plagioclase, opaques (Ti oxides), $\pm$ garnets, with minor quartz and apatite as accessories. They also contain good amount of brownish or greenish brown hornblende, which ranges from 20 to 40 modal percentage. The hornblende replaces the pyroxenes to different degrees. Further, the formation of late fibrous to platy anthophyllite replacing pyroxene along the fractures and grain boundaries via the reaction $\text{Opx} + \text{Qtz} + \text{V} = \text{Anthophyllite}$ is observed. Apart from amphibole brown to reddish brown biotite is also found to replace pyroxene via amphibole in these rocks. They are subdivided into the following types on the basis of mineral assemblage: (a) garnet bearing two-pyroxene granulites, (b) garnet absent two-pyroxene granulites, (c) garnet pyroxenites, (d) transitional amphibolite-pyroxene granulites, and (e) metadolerite dykes (Kaul *et al.*, 1985) with a granulitic mineral composition.

The mafic granulites are mostly equigranular, medium to coarse grained, showing xenoblastic granular texture. Some of the mafic granulites which occur as boudin preserve primary granulite facies assemblage. The texture is equigranular and polygonal with straight boundaries and the grains meet at triple point. Generally, the mafic granulites are non-foliated, however, the sheared

varieties impart a weak foliation due to alignment of amphiboles and biotite and elongation of pyroxenes and plagioclase. The disequilibrium assemblage of the mafic granulites indicate a superimposition of amphibolite facies metamorphism over the earlier granulite facies metamorphism. The post-granulite deformation has extensively affected the mafic granulites producing varieties of deformation microstructures and development of amphiboles and biotites. The increasing intensity of deformation can be correlated with the high modal percentage of amphiboles and biotites. The deformation microstructures indicate a high temperature environment of deformation.

Orthopyroxene (mainly hypersthene) occur as evengrained xenoblastic to granular aggregate with palegreen to pink pleochroism. In some sections they meet at triple point junctions indicating the recrystallized nature. In few cases orthopyroxene along with clinopyroxene occurs as coarse grained porphyroblasts. This may indicate that they belong to the primary mineral assemblage equilibrated under granulite facies conditions. Some of the orthopyroxene show very fine exsolution lamellae of clinopyroxene and these grains do not extinct completely. The contacts between orthopyroxene and clinopyroxene and plagioclase are mostly straight and in some cases having xenoblastic grain boundaries. In general, the orthopyroxene show undulose extinction and minor recrystallization.

The clinopyroxene is a primary constituent in textural equilibrium with orthopyroxene. It occurs as well developed to xenoblastic grain boundaries and often contains fine exsolution lamellae. Similar to orthopyroxene, the clinopyroxene also show crystal plastic deformation feature. Very rarely clinopyroxene contain small grains (one or two) of garnet, probably due to the result of exsolution. The other exsolution product is titanium oxide and plagioclase is absent.

Plagioclase occurs as even grains with xenoclastic to well defined straight grain boundaries. It is twinned but do not show any optical zoning. Most of the grains show undulose extinction indicating a superimposed moderate to strong deformation. The plagioclase-plagioclase contact is often polygonal and the grains meet at triple point junctions indicating recrystallization under higher temperatures. Their contact with pyroxenes are sharp with concave or convex grain boundaries.

The opaques are of two varieties, one the primary with well bounded crystal faces and the other is secondary with anhedral grains formed during conversion (hydration) of pyroxenes to amphiboles/biotites under lowering of temperature conditions.

The garnet is present as coronas around pyroxenes or at the contact between pyroxene and plagioclase. The garnet in these rocks, however, persisted during

the decompression event, in contrast to widespread evidence of its breakdown under similar conditions elsewhere (Harley, 1988, 1989). In some cases garnets occur as anhedral poikiloblasts containing numerous inclusions of plagioclase and pyroxenes. The coronatic garnets in some cases coalesce to form coarse grain garnets. These poikiloblasts and other coarse garnets, seem to have attained granoblastic shapes subsequently. In one case, the garnet lamellae on orthopyroxene- plagioclase contact continue to develop across a orthopyroxene grain probably along kink band boundary of orthopyroxene. The textural criteria indicate that garnet has developed by the reactions $\text{opx} + \text{plag} = \text{gar} + \text{qtz}$ and $\text{cpx} + \text{plag} = \text{gar} + \text{qtz}$.

Fluid Inclusion Studies

Fluid inclusion studies were carried out at Wadia Institute of Himalayan Geology, Dehradun, India on doubly polished thin rock wafers (about 0.2 to 0.5 mm). The petrography is carried out using high resolution microscope and the microthermometric runs were conducted on SGE gas flow heating-freezing system. The stage was calibrated using pure CO_2 inclusions, the Gonzals fluorite and Omegalaq temperature indicating liquid. The quoted values have a precision of $\pm 0.2^\circ\text{C}$, whereas for higher temperature the range fluctuated to about 1 to 2°C . The densities and isochores are calculated using Flincore Programme after Brown (1989).

Felsic Rocks of the Region

The inclusions observed represent the various stages of fluid processes. They are categorised into following types:

(i) **Type A inclusions** : These are monophase inclusions. Large number of such monophase inclusions filled with a dense carbonic liquid are observed in these gneisses. They develop a second phase upon cooling. These inclusions are more abundant in garnets than in quartz grains. The common size of these inclusions is 10 ± 5 micron. However, those with <5 micron and upto 20 micron are not exceptional. They are mostly regular with a variation from slightly irregular to equant. They occur in a variety of disposition patterns like isolated inclusions (Fig. 1a) which are more common in garnets or in small groups (Fig. 1b) or in small trails occurring along with carbonic aqueous inclusions (Fig. 1c).

(ii) **Type B inclusions** : Most abundant inclusions are type B biphasic which on cooling nucleate a third phase showing fluid immiscibility, consistent with the carbonic-aqueous composition. Although most of these inclusions are biphasic at room temperature yet in a few exceptional cases a third CO_2 gas

phase is also visible. They show wide variation in the size from <5 micron to about 20 or even upto 25 micron. They are subrounded, elongated, irregular to perfect equant in shape.

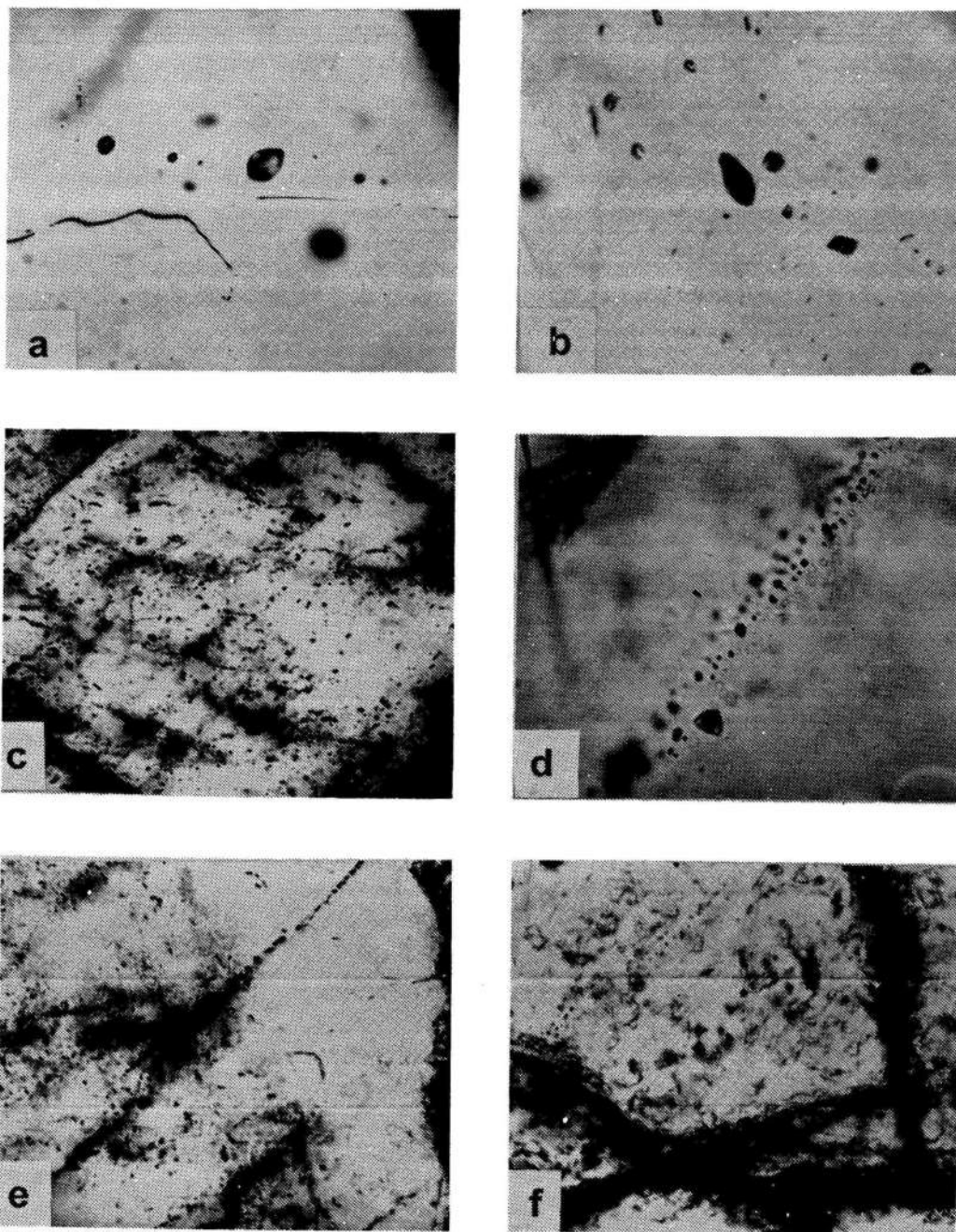


Fig. 1. Photomicrographs of fluid inclusions in felsic rocks, Schirmacher region, East Antarctica. Photographs were taken using plane polarized light. (a) Subequant isolated CO₂ inclusions (20 microns). (b) Group of CO₂ inclusions (<15 microns). (c) CO₂ inclusions in small trails (<10 microns). (d) Trail of CO₂ inclusions and CO₂-H₂O inclusions. (e) CO₂-H₂O (LCA) inclusions in healed cracks. (f) H₂O-NaCl (L-V) inclusions in secondary trails

Two types of CO₂-H₂O inclusions are observed, viz., (i) The Early Carbonic Aqueous (ECA) inclusions, and (ii) The Late Carbonic Aqueous (LCA) inclusions. The ECA inclusions represent an early CO₂-H₂O fluid phase and are found in small groups and as pseudosecondary trails associated with monophasic carbonic inclusions (Fig. 1d). They show an average 0.2 degree of fill consistent with about 80% CO₂ by volume. The second generation LCA inclusions (Fig. 1e) occur in secondary rehealed trails intersecting the grain boundaries and in planes associated with the biphasic aqueous inclusions.

(iii) **Type C inclusions** : These are biphasic aqueous inclusions found in healed cracks, often associated with late CO₂-H₂O inclusions. Besides a dominant liquid phase at room temperature, these inclusions also have a vapour bubble occupying about 10 to 40 volume % (Fig. 1f). Most of these inclusions are <20 micron in size and are subregular in shape.

The inclusions petrography suggest a sequential entrapment of various fluids. Monophasic carbonic inclusions are considered as earliest on the basis of their isolated occurrence and are followed by entrapment of monophasic carbonic inclusions in trails. LCA inclusions and the aqueous inclusions appears to be trapped at a late stage. The abundance of these inclusion types is not uniform exhibiting unsystematic distribution in the studied samples. Nevertheless, the predominance of LCA and H₂O- NaCl inclusions in some samples reflects H₂O rich fluid phase and presence of conduits for the circulation of such secondary fluid system.

Microthermometry and Density Estimates

The microthermometric estimates of monophasic inclusions show that melting temperatures are in the range of -56.2 to -57.8°C which nearly coincide with the melting temperatures of pure CO₂ compositions (-56.6°C). The little depression observed in these values may be because of minor amount of dissolved species like CH₄ and/or N₂. Homogenization of monophasic CO₂ inclusions vary from -12.5 to +16.2°C with peaks well constrained around -6 and +8°C. Further, resolution shows that — isolated carbonic inclusions are confined to a range of -12.5 to -0.8°C and those found in trails homogenized between +3.8 to +16.2°C. This homogenization data of isolated CO₂ inclusions is consistent with a CO₂ density range of 0.996 to 0.933 gm/cm³, while the densities estimated for the CO₂ inclusions in trails is relatively low varying between 0.905 to 0.811 gm/cm³.

The microthermometric measurements of ECA inclusions agree with CO₂-H₂O composition of the trapped fluid. The carbonic phase in these inclusions show melting at around -57.5°C and homogenization between +15.6 to +24.2°C with peak around +18°C. These temperatures are in agreement with a

CO₂ density of 0.816 to 0.723 gm/cm³. The clathrate melting could be observed in only a few of LCA inclusions, at an average temperature of +8.5°C which corresponds to a salinity of the aqueous phase as about 3 wt % NaCl equ. and suggest CO₂-H₂O-NaCl composition of these inclusions. The clathrate formation could not be observed in ECA inclusions. Furthermore, this temperature in LCA appears to be too low in comparison to the high salinity of the coexisting H₂O- NaCl inclusions (salinity upto 9 wt % NaCl equ.).

The microthermometric data of LCA inclusions indicate that the composition of fluid was changed at a later stage with influx of water. The homogenization of CO₂ into liquid phase in LCA inclusions range from +18.6 to +29.2°C. This is consistent with a CO₂ density between 0.788 to 0.624 gm/cm³. The total homogenization of LCA inclusions into gas phase took place at a temperature of 190 to 360°C.

In the aqueous inclusions first melting occurred at temperatures around -22°C which is close to H₂O-NaCl-KCl eutectic. The final ice-melting temperature ranges from -3.5 to -5.8°C, indicate low saline aqueous nature of these inclusions with nearly 5 to 9 wt. % NaCl equi. salts. These H₂O-NaCl inclusions homogenized between 210 to 300°C with a maxima around 260°C.

Fluid Phases Evolution

The fluid inclusions helped in identifying distinct fluid phases which range from CO₂ rich to H₂O dominated, with an intermediate wide regime of mixed CO₂-H₂O fluids. The CO₂ rich fluids are common both in metamorphic belts as well as in the deep seated intrusions. The influence of CO₂ rich fluid at the initial stage of metamorphism is considered on the basis of petrographic evidences for early trapping of CO₂ rich inclusions. But a wide variation has been observed in CO₂ densities which reflects varying P-T conditions of entrapment during cooling of these rocks. At this early stage molar volume of CO₂ was around 44.1 to 54.2. The densities of trail bounded CO₂ inclusions (0.902 to 0.801 gm/cm³) nearly match with coexisting ECA inclusions (0.816 to 0.723 gm/cm³) indicating simultaneous entrapment. This stage records first fluid immiscibility event when water was introduced to the fluid system. Considering the 0.8 degree of fill a composition as XCO₂=0.558 and XH₂O=0.442 is calculated for this stage fluid. The microthermometric data of LCA inclusions indicate that the composition of fluid was changed at a later stage with influx of water. LCA inclusions show lowest CO₂ density (0.625 gm/cm³) perhaps the vanishing stage of CO₂ in the fluid that had an average composition of XH₂O=0.767, XCO₂=0.225 and XNaCl=0.070. This calculation is based on the average clathrate melting temperature (+8.5°C) which could be observed only in few inclusions, the value may even be less than this noticed

value. The abundance and the association of CO₂-H₂O fluid (ECA and LCA) indicate wide span of mixed fluid activity in mineral fluid equilibria reactions. LCA inclusions coexist with H₂O-NaCl inclusions in secondary trails and this represent the record of compositional change in metamorphic fluid. The H₂O-NaCl fluid has salinity about 5.6 to 9 wt % NaCl equ. and bulk density as 0.735 to 0.936 gm/cm³ and shows an inverse relation of T_m and T_h. Higher the T_h lower is the salinity which imply for the coexistence of dilute fluid at upper P-T conditions and a gradual concentration towards lower P-T range or lower grade. H₂O-NaCl inclusions are secondary and the upper T_h of these inclusions is around 350°C suggesting late entrapment of this fluid.

From the overwhelming abundance of CO₂-H₂O inclusions it is apparent that the immiscible fluid had a dominating role in fluid related metamorphic history of these rocks as also observed in several studies on metamorphic rocks (Santosh, 1987 and references therein). The immiscibility may occur by mixing or unmixing of fluid components. Fluid mixing is a common phenomena which may result in the entrapment of CO₂-H₂O inclusions with wide range of fluid proportions. Such a mixing may be suggested for late H₂O-NaCl and LCA inclusions as these are found in rehealed fractures which adds in the mixing process by circulating fluids (Touret, 1977). However, the unmixing may occur from a homogenous fluid cooling from high temperature. As per the understanding of the volumetric properties of CO₂-H₂O-NaCl system the solvus crest for pure CO₂-H₂O system expands largely from 275°C at 1 kbar to higher P-T conditions with the addition of salts. Thus, even at higher P-T conditions immiscible carbonic aqueous saline fluid may coexist. But, for the present case of CO₂-H₂O ECA inclusions coexistence of unmixed fluid is not possible at high temperature. It can only be assessed that the fluid components might have been unmixed during uplift related pressure drop. But for the coexisting LCA and H₂O-NaCl inclusions the entrapment of unmixed fluid is possible because pressure which has adverse effect on CO₂-H₂O solvus was also low at this stage.

The continuum in the fluid composition from CO₂ to H₂O-NaCl imply for a slow mineral equilibria process as well as stress for the lowering of P-T conditions at successive stages. The densities and compositional change of the fluid argues for retrograde effects. Fluid inclusions have been widely used in defining such retrogression and P-T path (Hollister *et al.*, 1979; Lamb, *et al.*, 1991). The inferred retrogression in the presence of carbonic and saline aqueous fluid has also been reported in other areas. In some samples from the shear zones a relatively high abundance of H₂O rich fluid phase is observed which stress for higher retrograde effects. This may be because of large open surface to interact more with low temperature late fluids.

Geothermobarometry of Felsic Rocks

The understanding of phase volume and density data help to determine isochores reflecting P-T path along which the fluid was trapped. In the present case isochores of pure CO₂ inclusions suggest an initial pressures of about 6.8 kbar at 850°C. Further, the intersection of the isochores of peak density of ECA inclusions with the known density of coexisting CO₂ fluid infer the P-T conditions at which visible amount of water was introduced to the initial CO₂ rich fluid system. This event occurred at about 650°C and 4.5 kbar and represent an episode of changed metamorphic conditions. A compositional change from CO₂ rich fluid regime to H₂O rich fluid system is recorded between temperatures and pressures of about 350 to 450°C and 1.5 to 2.4 kbar. At this range LCA and H₂O-NaCl inclusions were trapped and most hydrous reactions in the rock occurred.

Mafic Granulites of the Region

The principal fluid phases observed in the mafic granulite sections are as follows:

(i) **CO₂ rich inclusions** : Primary monophasic inclusions containing a dense carbonic fluid phase are abundant in these samples. The liquid present completely fills the cavity and upon freezing, a gas bubble is found to develop in these inclusions. They resemble with the CO₂ inclusions reported from other high grade terrains. These inclusions occur in variety of distribution patterns, as isolated random distribution, in small groups, trails and sometimes in association with mixed fluids (Figs. 2a to 2c). Two main categories can be identified, viz., those found as isolated scattered fashion and other in planar arrays sometimes seen associated with carbonic aqueous inclusions (Fig. 2d). They are mostly <20 micron in size and subrounded, polygonal or subregular in shape.

(ii) **Carbonic-aqueous inclusions** : These biphasic inclusions at room temperature consist of a transparent water phase besides carbonic liquid. Upon cooling a third gas phase is developed in these inclusions thereby suggesting an aqueous-carbonic composition of the fluid. They commonly occur in trails and groups, at times associated with monophasic carbonic inclusions. CO₂-H₂O proportion estimated from relative volume of these phases indicate that there are two categories having 30 to 50 vol. % and other >70 vol. % CO₂ in inclusions. They are nearly identical to the mixed carbonic aqueous inclusions found in the associated leucogneiss (Rameshwar Rao and Rajesh Sharma, 1994). However, their size in granulite samples is small and population density

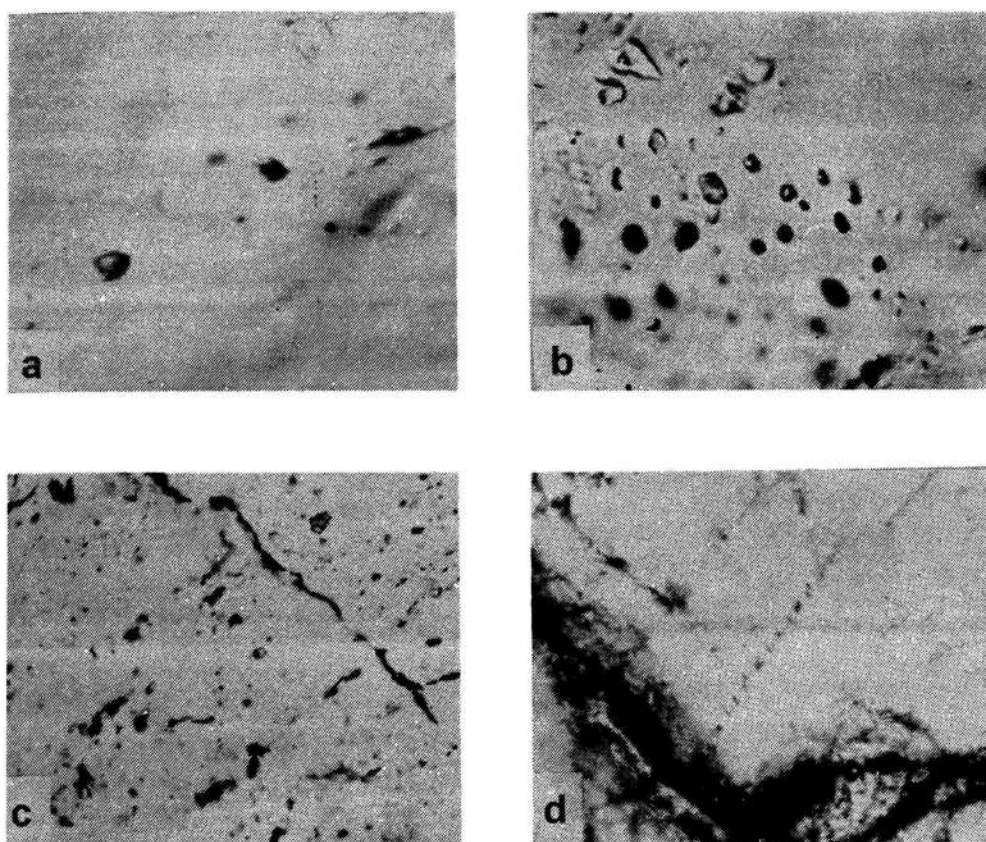


Fig. 2. Photomicrographs of fluid inclusions in mafic granulites, Schirmacher region, East Antarctica. (a) Isolated CO₂ inclusions. (b) Group of CO₂ inclusions. (c) CO₂-H₂O inclusions in trails. (d) Secondary H₂O-NaCl inclusions

is relatively low. They are generally 5-15 micron in size and subrounded, irregular or elongated in shape.

(iii) **Aqueous fluid inclusions** : Small (<10 micron) biphasic inclusions, subregular in shape, with liquid-vapour (water) composition are also present, but they are less abundant. The vapour bubble occupies 10 to 30 vol. % of inclusion cavity. They are distributed along late trails denoting a predominance of water rich fluid at a late stage. They are low saline as salt crystal is not found in them.

Order of Entrapment

Nearly all of the samples studied contain the above mentioned fluid phases, however, their proportion varies in different samples. Generally, the CO₂ rich inclusions form the most prominent variety. The density data of these inclusion types show a consistent variation. The petrography of inclusions suggest that isolated monophase CO₂ inclusions were trapped earliest, followed by cotrap-

ping of CO₂ and the CO₂-H₂O inclusions in planar arrays, indicated by their coexistence. Further, the aqueous fluids and low carbonic mixed fluid represent a late entrapment as evident from their occurrence in secondary trails. The petrographic studies of these rocks indicate the effects of retrograde metamorphism, and the mixed and aqueous fluids appears to have participated in these retrogressive event. Since the monophase CO₂ inclusions have textures indicating their early formation, they are considered as representative of peak metamorphic granulite phase and they are of fundamental importance for the current discussion and have been focussed here.

Results of Microthermometry

Microthermometry is primarily based on low temperature phase relations of carbonic system. Melting of solid CO₂ in isolated CO₂ inclusions occurred between -57.8 to -56.5°C and in trail bounded CO₂ inclusions in a range of -57.4 to -56.5°C, suggesting the presence of nearly pure CO₂ in these inclusions. The peak of CO₂ melting temperature near -57°C indicate that about 5 mole % or less CH₄ may be present if we consider that only CH₄ is responsible for this depression in melting point.

The overall homogenization temperature range varies from +13.5 to -25.4°C. Of this range the isolated primary inclusions were homogenized between -25.4 to -10.5°C with a well defined peak at around -21°C, whereas this temperature for trail bounded CO₂ inclusions varies between +4.6 to +13.5°C, showing a peak at about +11°C. The calculated molar volumes for the peak temperature values using Flincore programme (Brown, 1989) are 42.44 and 51.51 for isolated and trail bounded CO₂ respectively. This homogenization data of different samples shows that little fluctuation in temperature exists, suggesting fluid entrapment at varying conditions, however, these are in close range. The isochore calculated for the maximum density and for histogram peak (+11°C) of trail bounded CO₂ inclusions is at low isograde level. Though the density of CO₂ may be influenced by CH₄ but as it forms only <5 mol. % in the inclusion fluid, isochores are not much affected in the present case.

Mineral Chemistry

Mineral analyses were performed on the JEOL JCXA-733 super probe at University Scientific Instrumentation Centre, Roorkee, India. A probe current of 20 nA at an accelerating voltage of 15 KeV and a beam size of 2 microns (defocussed beam of 10 microns for feldspars) are used. Standardization was conducted against natural SPI multi-standards using ZAF corrections.

Microprobe analysis of the plagioclase do not show any compositional difference between core and rim. The An content of mafic granulites varies from 53 to 89, those of garnet biotite gneisses range from 43 to 56, orthopyroxene gneisses range from 30 to 34, while augen gneisses and leuco gneisses are around 31 and 21 respectively.

The enstatite end-member of orthopyroxene Mg% is high in mafic granulites ranging from 48 to 61%, while in orthopyroxene gneisses it ranges from 42 to 44%. They generally fall in the hypersthene composition of enstatite-ferrosilite solid solution series. The plotting of $\text{Al}_2\text{O}_3\%$ vs $\text{MgO}+\text{FeO}\%$ of orthopyroxene show that, they are metamorphic pyroxenes. The Mg% of clinopyroxene of mafic granulites range from 61 to 74% and represent clinohypersthene composition.

The analyses of garnets from the rocks of Schirmacher region do not show much zoning. However, the microprobe analyses of garnet from leucogneiss shows zoning. It is characterized by rim enrichment in spessartine, rim depletion in pyrope, almandine increase from core to rim and marginal decrease of grossular. The zoned garnet is considered to have Mg rich relict core of early metamorphism with an envelope of retrograde outer rim. It is also observed that the mafic granulites are relatively rich in Ca and Mg contents, while orthopyroxene gneisses and garnet biotite gneisses are relatively rich in Fe and Mn contents.

Compositionally, micas like plagioclase exhibit a narrow variation from core to rim, indicating no zoning. It is observed that the alteration of biotite to green biotite lowers Fe, Ti, Al and X_{Fe} (*i.e.*, Fe/Fe+Mg) contents.

P-T Conditions

Recently, numerous workers have calibrated cationic exchange equilibria and net-transfer equilibria among co-existing minerals as a function of temperature and pressure in an effort to develop practical methods of inferring the P-T conditions of metamorphosed rocks. Based on the mineral assemblage of the rocks, different calibrations have been applied to deduce the P-T conditions of rocks. For geothermometry mineral pairs like orthopyroxene-clinopyroxene; garnet-orthopyroxene; garnet-clinopyroxene, biotite-garnet, alkali feldspar-plagioclase feldspars have been used. For biotite-garnet thermometry and alkali feldspar-plagioclase geothermometry the software programs of Rameshwar Rao (1995) and Rameshwar Rao and Subba Rao (1994) are used. The pressure estimates were made using mineral assemblage of garnet-pyroxene-plagioclase-quartz and garnet-muscovite-biotite-plagioclase-quartz. For each indi-

vidual set of coexisting minerals 3 to 8 different calibrations given by different workers are used before calculating the mean values.

Overall, two major episodes of metamorphism are characterized using P-T data, one the early metamorphism recording higher temperature and pressure conditions and the other subsequent latter retrograde metamorphism of relatively lower temperatures and pressures. The two episodes of metamorphism are observed in mafic granulites and orthopyroxene gneisses.

The orthopyroxene gneisses recorded temperatures and pressures of $650\pm 50^{\circ}\text{C}$, 5.7 ± 0.4 kbar and $760\pm 75^{\circ}\text{C}$, 7.0 ± 0.7 kbar. The garnet biotite gneisses recorded temperatures of $610\pm 50^{\circ}\text{C}$ and pressures of 5.8 ± 0.5 kbar, while augen gneisses recorded temperatures of $625\pm 50^{\circ}\text{C}$ and pressures of 5.7 ± 0.5 kbar. The garnet rim and biotite matrix thermometer gave lower temperatures, in the range of 648 to 728°C , with a mean value of $675\pm 50^{\circ}\text{C}$ and pressures of 5.5 ± 0.6 kbar. The temperatures obtained range between 434 to 456°C with a mean value of $444\pm 12^{\circ}\text{C}$. The results obtained are lower than those obtained by garnet (rim)-biotite (matrix) geothermometry. These low values indicate the temperatures at which re-equilibration of the participating feldspar phases was attained.

The mafic granulites recorded two groups of temperatures. One group is shown by the primary orthopyroxene-clinopyroxene mineral assemblage which retain the higher temperature signatures along with garnet-clinopyroxene (core) pair. Though, we do agree that the granulite facies temperatures are not well preserved due to resetting, within the limitations of available data we propose a best estimate temperature for the granulite facies metamorphism of around $810\pm 50^{\circ}\text{C}$. This includes the best estimates for the metamorphic temperatures such as Ellis and Green (1979) and Krogh (1988) and also values from orthopyroxene-clinopyroxene thermometry after attributing necessary corrections for the uncertainties. The second group of temperatures from these rocks are averaging around 660°C with a standard deviation of around 35°C , which covers 80% of garnet-biotite geothermometers (like Ferry and Spear, 1978; Perchuk and Lavent'eva, 1983; Indares and Martignole, 1985; Bhattacharya *et al.*, 1992), garnet-orthopyroxene geothermometers (like Harley, 1984 and Sen and Bhattacharya, 1984), garnet-clinopyroxene geothermometers (like Ellis and Green, 1979 and Krogh, 1988) and garnet-hornblende geothermometers (like Graham and Powell, 1984). The pressure estimates with garnet-pyroxenes-plagioclase-quartz (with clinopyroxene core composition) at a preferred temperature of 1023°K using calibration of Perkins and Newton (1981); Newton and Perkins (1982); Bohlen *et al.* (1983); P-Fe calibration of Bhattacharya *et al.* (1991) and Eckert *et al.* (1991) indicate a pressure range of 6.6

to 7.6, with a mean value of 7.1 and standard deviation of 0.3 kbar. The other group is characterized by pressure estimates at a preferred temperature of 973°C, with (i) the garnet-pyroxenes-plagioclase-quartz assemblage using the calibrations of Perkins and Newton (1981); Newton and Perkins (1982); Bohlen *et al.* (1983); Bhattacharya *et al.* (1991) and Eckert *et al.* (1991), (ii) the garnet-biotite-plagioclase-quartz assemblage using the calibration of Hoisch (1990) and (iii) the garnet-hornblende-plagioclase-quartz assemblage with clinopyroxene rim composition, using the calibration of Kohn and Spear (1990). The estimated pressures fall within the range of 5.1 to 6.0 with a mean value of 5.7 and standard deviation of 0.3 kbar.

Petrogenesis

A careful documentation of the P-T conditions combined with characteristics of fluids and their tectonic implications in the felsic gneisses and mafic granulites are discussed in order to understand the evolution of metamorphic terrain in the Schirmacher region, East Antarctica.

P-T Conditions and Fluid Evolution of Felsic Gneisses

The P-T mineral equilibria obtained with garnet (rim) computes $675 \pm 50^\circ\text{C}$ and 5.5 ± 0.6 kbar, more or less coinciding with changed metamorphic event wherein introduction of water to CO_2 rich fluid system has taken place (650°C and 4.5 kbar). However, the upper pressure range estimated from mineral equilibria is higher than the pressure determined from peak CO_2 density. Such difference in estimated pressure may be because of invisible water in CO_2 fluid inclusions (Lamb *et al.*, 1991). A second possibility of such difference could be because of modification in inclusion during uplift and changed P-T conditions (Sterner and Bodnar, 1989). Further, the compositional change of system to H_2O dominating fluids resulting in the formation of most hydrous minerals in the rock (350 to 480°C and 1.5 to 2.4 kbar) coincides with the temperatures of re-equilibrium in the feldspar phases ($444 \pm 12^\circ\text{C}$).

The influence of CO_2 rich fluid at the initial stage of metamorphism is considered on the basis of petrographic evidences for early trapping of CO_2 rich inclusions. But a wide variation has been observed in CO_2 densities and isochores (6.8 kbar at 850°C), which probably reflects varying P-T conditions of entrapment during cooling of these rocks. The close range of P-T conditions obtained from these methods suggest that the fluids were trapped at the peak metamorphic conditions. The ECA inclusions coexisting with monophase CO_2 inclusions in trails and their microthermometric data, records the compositional

change in metamorphic fluid at a late stage in the evolution of rocks from earlier granulite facies metamorphism with influx of water.

P-T Conditions and Fluid Evolution of Mafic Granulites

The pressure and temperature conditions of mafic granulites of Schirmacher region range from nearly 7.6 to 5.1 kbar and 860 to 600°C respectively, but it needs to be mentioned here that no clear correlation with rock type or mineral composition is observed. This is because, the mineral compositions used for P-T estimates may never have been in equilibria with each other or a later equilibration affected different minerals to different extent. Thus, it is observed that the mineral assemblage register the resetted amphibolite facies conditions representing the re-equilibration between coexisting phases on cooling towards blocking temperatures for cation exchange. However, near peak granulite conditions are recorded by zoned clinopyroxene minerals and by primary mineral assemblage of orthopyroxene-clinopyroxene. The composition of An₈₄ and An₇₂ in two of the samples studied also suggest the partial preservation of the early granulite facies metamorphism in these rocks. The subsequent uplift and cooling of the region has continuously re-adjusted the minerals to lower P-T conditions. This event as culminated in a widespread major deformation event, that represents the amphibolite facies metamorphism observed in the Schirmacher region. The summarized two groups of P-T data in earlier section correspond respectively to the earlier granulite facies metamorphism (810±50°C and 7.1±0.3 kbar) and latter retrograde amphibolite facies metamorphism (660±35°C and 5.7±0.3 kbar) that followed the decompression path.

The proposed values compare favourably with the estimates based on fluid inclusion studies for peak densities. The change in grade of early granulite facies and latter retrograde amphibolite facies metamorphism is indicated by the increasing dominance of CO₂ over H₂O in the early fluid phases. The density of isolated CO₂ inclusions range from 0.986 to 1.057 gm/cm³ with a histogram peak at 1.037 gm/cm³. This value denote the conditions at which maximum entrapment took place. The isochores for early CO₂ inclusions penetrates the P-T box of metamorphic conditions deduced from mineral composition. A pressure of about 7.1 kbar at 800°C is inferred from the isochore for histogram peak value (density=1.037 gm/cm³). This entrapment pressure match with mineral phase barometry thus suggesting that CO₂ inclusions were trapped during peak metamorphic conditions (Lamb *et al.*, 1987). Consequently, the fluids trapped in early carbonic inclusions present in garnet and quartz are considered to represent peak metamorphic fluids (Touret, 1988).

The density has also been estimated for the trail bounded CO₂ inclusions in which H₂O was optically undetectable. These values are at a lower side varying from 0.899 to 0.835 gm/cm³. Density of their histogram peak is at 0.854 gm/cm³. The low density of these CO₂ inclusions and the entrapment of mixed H₂O-CO₂ inclusions, post dating the peak granulite metamorphism is attributed to regional retrogression and hydration event which has also been inferred through the mineral phase thermobarometry.

It is to be noted that, most samples do not show the complete temperature range and are found to preserve unequal distribution of the density range. This signifies that the peak granulite conditions are not well preserved in all the samples and the effects of retrogression are also not uniformly imposed. The variation and lowering in densities could also be because of partial reequilibration of fluids at lower P-T conditions during uplift (Sterner and Bodnar, 1989).

Emplacement Depths, Crustal Thickness and Geothermal Gradients

Assuming a geobaric gradient of 300 bar/km, the P-T conditions of the leucogneiss corresponding to garnet (rim) composition indicate around 18 km of the crust to have been removed. If we consider that the event where water appeared in the system represents the near melting conditions, the highest density recorded from trail bounded CO₂ inclusions may be used to understand the depth conditions. Thus, following the approach of Belkin, *et al.* (1985), the CO₂ density estimate of 0.905 gm/cm³ at about 700°C infers a depth of about 17 km. Restoration of eroded material suggest post-emplacement crustal thickness of >50 km. Based on these depths and temperatures estimated, geothermal gradients of atleast 37°C/km during amphibolite facies metamorphism are inferred.

The pressures recorded in the mafic granulites correspond to a crustal depth of 19-24 km, this added to the assumed present day crustal thickness of nearly 35 km, indicate a pre-erosional crustal thickness of 54-60 km. The deduced P-T conditions require an average geothermal gradient of around 35°C/km implying minimum heat flow of 100 mW m⁻². The tectonic regime of thickened crust (54-60 km) and elevated geothermal gradient (35°C/km) of the Schirmacher region link the evolutionary path to crustal sequence thickened by collisional overthrusting (Bohlen, 1991).

Granulite Genesis

The apparent absence of water in early CO₂ inclusions in the mafic granulites of the Schirmacher region, indicate that the fluid phase interacted with the granulite was anhydrous. This implies for low $a_{\text{H}_2\text{O}}$ at peak metamor-

phic stage (Newton, 1986), and that the CO₂ was the dominant volatile component in granulite stage fluid inclusions, like in many other granulite terrains (Touret, 1981). The abundance of primary CO₂ inclusions in peak metamorphic minerals (notably garnet) and the general correspondence between fluid and mineral data in the rocks under study confirm a 'fluid present' carbonic regime for the high-grade metamorphism. Although with present data and in the absence of carbon isotopic data the source of CO₂ is difficult to be deduced, a logical inference is however made for the studied rocks.

Source of Carbonic Fluids

A few sources have been proposed for the various granulite terrains worldwide. The main include (i) decarbonation reactions in deep crust (Glassley, 1983), (ii) graphitic reactions with hydrous silicates (French, 1966), (iii) release of CO₂ from deep-crustal granulite fluid inclusions, (iv) residual pore fluids in dehydration melting, (iv) convective recirculation, (v) CO₂ released from crystallizing magmas, and (vi) CO₂ streaming from mantle which enhance metamorphic reactions by lowering $a_{\text{H}_2\text{O}}$ and supplying heat (Janardhan *et al.*, 1979).

The general absence of graphite bands and the calc-silicate rocks in the mafic granulites of the Schirmacher region (except for some sporadic occurrences in the central and in small pockets in the eastern regions) suggest that, they do not form the main source in the formation of the granulites and may be considered as a local source for CO₂ fluids. The small quantities of CO₂ normally present in surface waters may not be sufficient to contribute substantially to deep metamorphism to support the convective circulation theory. In case of release of CO₂ from deep-crustal granulite fluid inclusions, it is difficult to account for abundant CO₂ in early deep-seated granulites. The mafic granulites of Schirmacher region are derived from mantle magmas as evident from their isotopic data (Sri <0.704), suggesting a mantle rich CO₂ source. Further, the possibility of CO₂ streaming from the mantle as suggested by Newton *et al.* (1980) is apparent from the interlayered relation of mafic granulites and orthopyroxene gneisses (charnockites) and the existence of CO₂ rich inclusions in small groups. The mantle reservoir contain free CO₂ fluid in their source. These fluids may be exolved on rising to lower crustal levels and get trapped as inclusions during tectonomagmatic episodes. Thus, it is viable to consider the influx of CO₂ bearing vapour plumes rising from upper mantle or CO₂ released from crystallizing magmas like some basalts that require entrapment of CO₂ rich fluids from mantle, as potential source for carbonic fluids in the Schirmacher region. A similar sub-continental mantle

source for CO_2 inclusions in the charnockites of Lutzow-Holm Bay region, East Antarctica has been suggested by Santosh and Yoshida (1992).

Retrograde P-T Trajectory and its Tectonic Implications

The fluid isochores have been widely used in suggesting the uplift path. A detailed account for different uplift paths is given by Harley (1989) and Lamb, *et al.* (1991), while Touret (1988) has discussed the conditions which support ITD and IBC retrogression paths. In the rocks under study, it is found that the mineral P-T estimates agree well with the results derived from fluid inclusions. Those isochores of CO_2 inclusions which penetrate the two P-T boxes may suggest the partial re-equilibration of the fluid at latter episode (Fig. 3). However, the

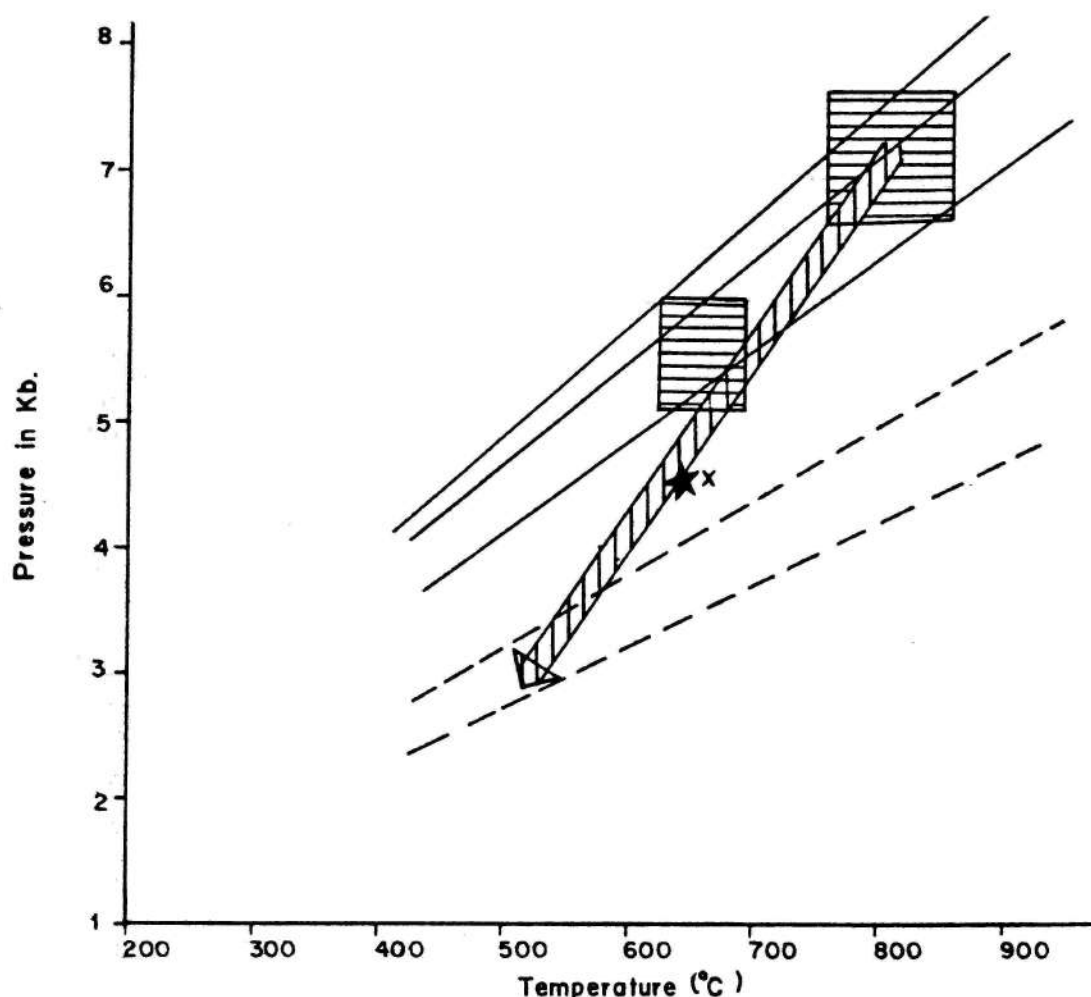


Fig. 3. P-T diagram showing selected CO_2 isochores for isolated inclusions (solid lines) and those in trails (dashed lines) with P-T range deduced from mineral thermobarometry (boxes). The point 'X' indicate the intersection point of isochores for coexisting CO_2 and $\text{CO}_2\text{-H}_2\text{O}$ inclusions observed in associated leucogneiss. The thick shaded arrow is the retrograde P-T path suggested for these granulites

isochores of late CO₂ inclusions are at low isograd level and many of our observations including mineral P-T estimates and the nature of fluid match with the ITD conditions described by Touret (1988). A dP/dT of 1.2 kbar/100°C without a vertical decrease in pressure is observed in these rocks. We have also recorded a P-T of about 4.5 kbar at 650°C for intersecting isochores of coexisting CO₂ and CO₂-H₂O inclusions in associated leucogneiss (Rameshwar Rao and Sharma, 1994). A decompressional P-T path is thus envisaged in these rocks, which pass through the equilibration conditions of 810°C and 7.1 kbar to 660°C and 5.7 kbar. The calculated segment of the P-T path is similar in slope and magnitude to decompressional path of Reinbolt Hills, East Antarctica. A near comparable situation is also observed in other Late Proterozoic granulite terrains of East Antarctica such as Rauer Islands, Larsemann Hills, which record decompression with variable amounts of uplift, deformation under amphibolite facies conditions.

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